

“Emotion Classification of Children’s Facial Expressions Using a CNN-LSTM Deep Learning”

A Dissertation

By

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**Department of Computer Science and Engineering
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A dissertation Submitted to the Department of Computer Science and Engineering,
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DEDICATION

I wholeheartedly dedicate this project to my beloved parents, whose endless love, encouragement, and sacrifices have been the foundation of my journey. To my respected supervisor and inspiring teachers, thank you for your continuous support, guidance, and faith in me through every step of this work.

Abstract

It can be hard to know how kids feel, but if we do, we can develop systems that are emotionally savvy and can make learning, mental health, and even computer gaming better. Adults can tell how someone is feeling by looking at their face, but it's usually easier to read kids' emotional reactions than it is for adults. The research suggests that a hybrid deep learning model that takes both space and time into consideration can help with this issue. We can use a Convolutional Neural Network unlawfully to see how expressions change over time in sequences and learn a lot about the space from facial pictures. The CNN-LSTM combination model is based on a precise set of children's facial expressions that show six emotions: happy, melancholy (the sense of good or unpleasant emotions), anger, fear, surprise, and neutrality. Based on the findings of trials, this technique is better and more reliable than the LSTM model and CNN (clinic model for human behavior) in a number of samples. There is proof in the real world that the claim is true. The technique typically works, but it has difficulties with confused faces, lousy lighting, and a lot of distinct emotional expressions. This plan isn't flawless, though. The model is meant to work without any extra hardware, though, so it may be utilized in places with minimal resources, like schools, telemedicine, and mobile apps. The results reveal that there are numerous conceivable uses for them, such as affect-aware" learning settings, mental health monitoring, smart tools, and interactive platforms that are safe for kids. This study reveals that automated facial expression analysis is a quick and uncomplicated way to find out how youngsters see themselves.

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I look forward to the possibility of continuing our collaboration in the future and remain sincerely appreciative of their support throughout this thesis.

Declaration of Originality

I certify that except where due acknowledgement has been made, the work is that of the author alone; the work has not been submitted previously in whole or in part to qualify for any other academic award; the content of the thesis is the result of work that has been carried out since the official commencement date of the approved research program; any editorial work, paid or unpaid, carried out by a third party is acknowledged; and ethics procedures and guidelines have been followed.

Author
July,2025

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Chapter One

Introduction

1.1 Introduction

Children in the present world, for instance, have different possibilities to see any video content online irrespective of its nature being acceptable or not. [1] Children in the modern world, for instance, have various options to view any video content. Features that in themselves may not be prevented in any fashion owing to other classic blockage strategies, such as keyword blocking or age blocking, would captivate the young people emotionally and mentally. More recently, there has arrived technology such as facial recognition systems that directly examine children and their potential to impact content heard or watched in real time. These systems use face recognition to detect symptoms of fear, discomfort, or panic and avoid the injury by either interrupting the video or telling the caretakers online, irrespective of its nature being appropriate or otherwise. One of the measures that will enable the achievement is the formulation of the model that will propose a proper classification of children's facial expressions since these signs differ from adult's emotional expression both in terms of look and perception. Currently, most of the facial recognition systems are developed based on adult faces; therefore, a high percentage of wrong results are expected for children's faces. Therefore, there is a need to develop a specialized model focused on the identification and categorization of children's emotions.

This study looks at the big differences in how children and adults disclose information. It presents a model for kids' facial affection confession and reviews the current literature on models that are based on adult-centered and child-centered strategies, showing how they are both ineffective and powerful. The method described involves creating a new convolutional neural network that can tell the difference between happy and sad faces. We did eight tests to see how well the models worked by changing the regularization and the animus appraisal process. Some of the key criteria we looked at were the right parade and affirmative loss anarchy matrices and how easy it was to understand the models using Grad-CAM visualizations.

1.2 The Necessity for a Child-Centric Facial Expression Recognition Model

Facial expressions alter a lot as people get older. Howard et al[2] say that young people and older people show very different expressions. These discrepancies present challenges for emotion detection systems, which are generally trained on photos of adult faces. Because of this age gap, these models don't work as well, Sayin and Aksoy[3] also argue that there isn't much research on how children and adults display emotions. It's to use adult-based emotion recognition methods on kids because their emotions change a lot due to growing up, developing mentally, and forming personalities.

Guodong Guo et al [4] looked at how aging impacts facial expression recognition. They identified huge differences between children and adults. They noted that kids utilize more distinct and forceful face motions, but adults employ quieter, smoother muscle movements because of their skin and wrinkles. Houstis and Kiliaridis [5] added that adults, especially men, show more vertical movement in their expressions, while children have less developed muscles, leading to different expression features across age groups. Guo also mentioned cultural issues, adding that these variations need to be explored to improve facial expression detection systems across all ages. Rao et al [6] investigated how expressions are identified in children and adults. They found that children's expressions are usually weaker and less clear, which leads to lower accuracy in recognition models no matter how many facial points are used. Another aspect that impacts emotional presentation is culture and ethnicity. For example, Lewis et al[7] observed that Japanese youngsters display less shame, pride, and sadness but greater humiliation compared to American kids. Camras et al [8] have noticed that Chinese babies smile and cry less than European American and Japanese babies. This research reveals that culture plays a large role in how emotions are displayed, making emotion recognition models even more complex.

1.3 Objective

A specific aim of this thesis is to develop a deep learning model capable of accurately classifying children's facial expressions, particularly joyful and sad emotions. To do this, the study intends to discover the key differences between children's and adults' facial expressions and investigate how these variations effect the accuracy of emotion detection systems. We will implement a hybrid CNN-LSTM model which is able to capture both the spatial and the sequential features of children's facial emotions. The model will be trained and tested with the "Happy and Sad Child Face Expression" dataset. Improving accuracy, consistency, and generalization will also remain our objectives. In addition, several experiments will be conducted with different regularization strategies and attention integrations in order to optimize model performance. To evaluate model explainability. We will apply visual techniques such as Grad-CAM to evaluate the interpretability of the model's decisions. Finally, the thesis will look at how this model is really employed in the real world, such as in child-safe media systems, smart education platforms, and child-computer interface technologies that can sense emotions.

1.4 Literature Review

Research on identifying children's emotions through facial expression analysis is still in its infancy, particularly since the majority of models currently in use were developed for adults and are ineffective for children. Recent studies have attempted to develop more accurate models designed with children in mind. Nagpal and their team [9] came up with a new model called the mean supervised deep Boltzmann machine msDBM. They tested it on two sets of face images and found it worked better than other models it scored 75.0% on one dataset and 48.0% on another, showing that models made for kids can be more effective.

Howard and others [10] looked into how cloud-based systems that identify emotions often work better for adults than for children. They created a better way to classify emotions by using existing models and custom ones tailored for kids. This method improved accuracy by 17.3% and helped the system perform better with less common emotions. Lopez-Rincon [11] tested two methods to recognize emotions in real-time using a robot called Nao. One method used software called AFFDEX-SDK, and the other was a combination of a CNN and another method called Viola-Jones. While CNN worked better overall, both had issues when dealing with emotions like disgust. Using both methods together improved accuracy to 46.05%, showing that combining different methods can help in real-time tasks. You [12] suggested a model that uses both facial movements and other signals like sounds to make the data more balanced. They used a new way to describe faces and a special method to select data. The model worked well on several datasets, achieving 97.8% accuracy on one, which shows that combining different features is very effective for recognizing emotion. Sayin and Aksoy [13] studied children's emotions by collecting face images from various sources and grouping them into seven types. They tested six models, including VGG16, ResNet50, and InceptionV3. InceptionV3 did best with 76.3% accuracy and a high F1 score; however, all models had trouble with emotions like fear and surprise, pointing to the need for more varied data and models that are better suited for children. Witherow [14] looked into using transfer learning to improve emotion recognition for kids. They first trained a model on adult data and then fine-tuned it with child data. While the model worked well on adult data, it improved significantly when trained specifically for children. They suggest that adult models can be adapted for kids, but there's still room for improvement through more diverse data and better model designs. Banerjee and others [15] developed models for children with autism that can work on mobile devices. They used a model called MobileNetV3-Large and tested it on a smartphone. The best model performed well, showing that training with child data helps the results. The results also showed that there are still differences based on race, showing the need for more inclusive data. Using a CNN with an auto encoder, Talaat [16] developed a real-time emotion identification system for kids with autism. Joy, anger, sadness and surprise were among the emotions used to test the model with a high AUC score and AUC accuracy of 95.23 percent. Exceptions outperformed the others. The system's potential for practical use is demonstrated by its utilization of fog computing and IoT using characteristics including texture and machine learning techniques. Rani (2017) investigated emotion perception *in children* with autism. The SVM model fared *better than the* neural network, demonstrating *that* conventional techniques can still be effective in limited datasets when combined with powerful features. s. Beiger and colleagues [18] compared children with autism to children with typical development in order to study how autistic children process emotions. According to the study, autistic kids reacted equally in social contexts but were less receptive to emotional cues in neutral environments. s. This implies that systems that consider context can help autistic youngsters participate emotionally. grammar check emotion recognition system for autistic children using a CNN and an auto encoder. The model was tested on emotions like joy, anger, sadness, and anger and sadness. Exceptions performed best, achieving 95.23% accuracy and a high AUC score. The system's use of fog computing and IoT shows its potential for real-world use. Rani [17] looked at emotion recognition in autistic children using features like texture and machine learning models.

The SVM model outperformed the neural network, showing that traditional methods can still be useful when paired with strong features in small datasets. Begeer and others [18] studied how autistic children process emotions by comparing them to typically developing kids. The study found that autistic children were less responsive to emotional cues in neutral settings but behaved similarly in social situations. This suggests that systems that take context into account can support emotional engagement in autistic children.

1.5 Problem Statement

In the field of affective computing, it is still very difficult to read and comprehend children's emotions from their facial expressions. The majority of contemporary facial expression recognition systems are largely trained on adult data. However, children's facial emotions differ widely in terms of look, strength, and change. These distinctions make adult-based models unreliable and prone to errors when applied to children. Therefore, there is a tremendous need to build emotion recognition algorithms customized to the face traits and behavioral patterns of youngsters. It may be more difficult to understand children's sentiments or their expressions than to understand their mental processes. Understanding the intricacies of how emotions evolve is a component that is absent from child and face readings, in addition to the holes left by traditional AI. As they notice facial qualities like eyes and noses, their thinking, emotions, and sentiments gradually and deliberately alter. A new strategy that takes into account both facial features and the way expressions evolve over time is necessary to boost the accuracy of emotion identification.

It is tough to progress systems that concentrate on children's emotions thanks to a lack of specialized models and high-quality datasets. Resolving this problem is crucial to developing intelligent systems in fields such as education, healthcare, entertainment, and online safety, where the ability to interpret a child's emotions can contribute to more responsive and empathetic computer interactions.

Chapter Two

METHODOLOGY

2. METHODOLOGY

2.1 Image Collection

We looked for copyright-free pictures of kids on the internet to generate a good dataset for sorting their facial expressions. Most of the time, the focus was on two basic feelings: happy and sad. There were 180 pictures in all. Of those, 100 were labeled as "Happy" and 80 as "Sad." Two randomly chosen people gave each image a second evaluation to make the classification process work better. At this stage, the photos were re-categorized based on the most prevailing emotional expression. This step was critical to guarantee that each image was appropriately sorted into its respective category. Figure 1 below shows a few representative photos from the collection.



Figure: 1 sample data from online

2.2 Dataset

This study uses the Happy and Sad Child Face Expression dataset, extended by adding 800 happy and 800 sad child face images, resulting in 1600 labelled .jpg images split evenly between happiness and sadness. The images were captured under varying lighting conditions and facial angles to support robust model training. Each image is labelled and stored in .jpg format, suitable for deep learning frameworks. The dataset is relevant as it focuses specifically on children's emotional expressions, which differ significantly from adults and are often underrepresented in standard datasets. Pre-processing steps included resizing, normalization, and data augmentation (flipping, rotation, brightness adjustment). The data was divided into training, validation, and test sets for unbiased evaluation. Limitations include having only two emotion classes, a small sample size, and variations in image quality, which may affect model accuracy.

2.2 Hybrid CNN-LSTM architecture

The proposed solution uses a mix of CNN and LSTM models to detect emotions on children's faces. The CNN part looks at each image frame and identifies important details like where the eyes are looking, the shape of the mouth, and key facial features. These details are then passed to the LSTM layer, which tracks how expressions change over time, helping the model understand fast emotional changes better.

Once the model is trained, the Adam optimizer is used to make it perform better, and categorical cross-entropy loss is used to measure how well it's doing. To prevent the model from remembering too much from the training data, techniques like dropout and L2 weight decay are applied. The model also uses attention mechanisms to focus on the parts of the face that are most important for emotions. To explain how the model makes its decisions, Grad-CAM is used to show which areas of the face were most influential in determining the final result.

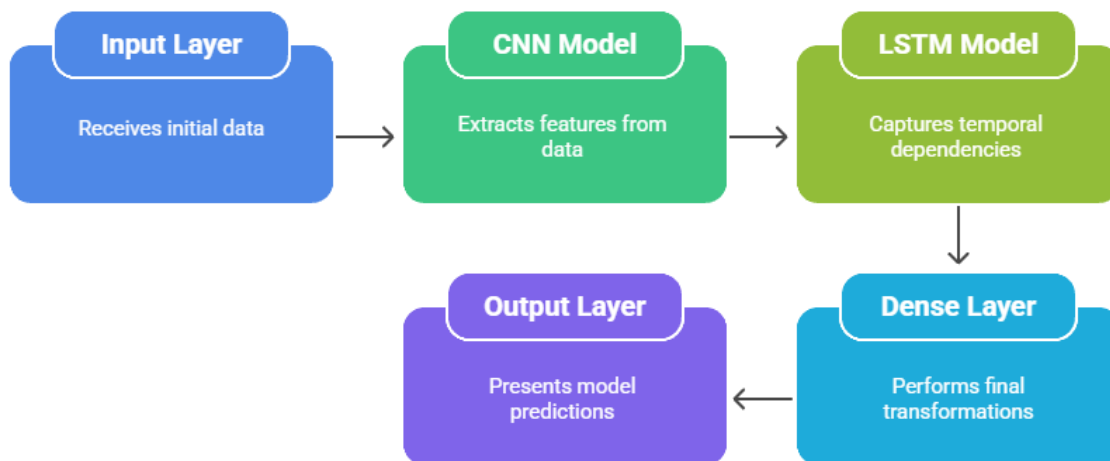


Figure 3: CNN-LSTM architecture

2.3 Generative Adversarial Network (GAN)

Generative Adversarial Networks (GANs) help machines to create new, realistic data by learning from existing examples. It is introduced by Ian Goodfellow and his team in 2014 and they have transformed how computers generate images, videos, music and more. Unlike traditional models that only recognize or classify data, they take a creative way by generating entirely new content that closely resembles real-world data. This ability helped various fields such as art, gaming, healthcare and data science. In this article, we will see more about GANs and its core concepts.

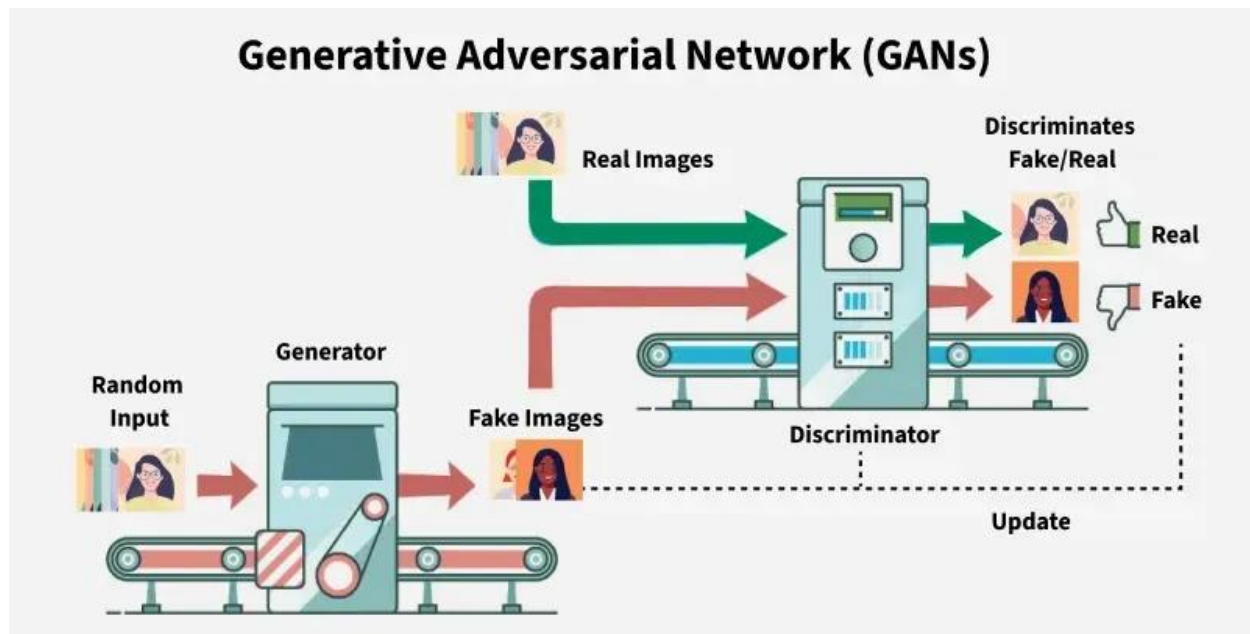


Figure 2: Generative Adversarial Network (GAN) Architecture.

Chapter Three

Expected Outcomes

3 Expected Outcomes

The proposed CNN-LSTM deep learning model is expected to greatly improve the ability to recognize children's facial expressions by using both spatial and temporal features. Compared to traditional models that use only CNN or LSTM, this combined approach will be better at capturing small emotional changes in children's faces, especially when expressions change over time. The model will be trained and tested by the "Happy and Sad Child Face Expression" dataset, and it is expected to show better classification results, be more reliable under different lighting, angles, and expression strengths, and work well with new, unseen data. The model will also use techniques like regularization and attention mechanisms to improve its performance by preventing it from learning too much from specific examples and focusing on the most important parts of the face. Tools such as Grad-CAM will be used to understand how the model makes decisions, which will make its predictions more transparent and trustworthy. If this model works well, it could be used in real-life situations like intelligent tutoring systems, child-friendly media, healthcare tools that detect emotions, and educational tools that respond to emotions. The overall goal of this study is to improve the understanding of emotions in children and help create safer, more sensitive interactions between children and computers.

3.1 Conclusion

Compared to adult-oriented platforms, developing a specialised model for categorising children's facial expressions fills a significant gap by improving emotion recognition by concentrating on how kids convey the "Happy" and "Sad" emotions. Along with robust data augmentation, the model's performance has been enhanced by sophisticated techniques, especially attention mechanisms like Squeeze-and-Excitation blocks and Convolutional Block Attention Modules. This was made possible through stable diffusion which led to the diversification and expansion of the dataset as well as the development of more realistic training samples.

Batch Normalisation with both Dropout and SE Attention achieved the highest accuracy, which proves just how effective these techniques are in improving generalisation. Thus, to follow the model's potential, additional advancements are still needed to provide the same or higher accuracy than models like MobileNet and ResNet18, extend the range of classification emotions, reduce the frequency of misunderstandings, and diversify and enhance the developed synthesised images. Future studies into more complex frameworks and training methods will be essential to tackling practical issues and creating a universal tool to safeguard kids' emotional health on the internet. Additionally, our work can be extended to other domains such as renewable energy [19] where data manipulation is required before training models [20].

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